

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8.	(a)	(i)		nitrogen forms NCl_3 ; phosphorus forms (PCl_3 and) PCl_5 (1) phosphorus is able to expand its octet but nitrogen is not / phosphorus has <u>available</u> <i>d</i> -orbitals whilst there are none in the outer shell of nitrogen (1)	2			2		
		(ii)		aluminium has three outer shell electrons so it is able to form three covalent bonds giving six outer shell electrons or an incomplete outer shell or leaving it electron deficient (1) dot and cross diagram of suitable example e.g. AlCl_3 and dot and cross diagram of suitable example that has formed coordinate bond e.g. AlCl_4^- / Al_2Cl_6 (1) DO NOT ACCEPT IONIC COMPOUNDS; COORDINATE BOND MUST BE CLEAR	2			2		